

FORT CHIPEWYAN, AB, Canada

WMO: 710560

Lat: 58.7664N

Lon: 111.1167W

Elev: 238

StdP: 98.50

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-36.4	-43.5	0.1	-38.9	-40.9	0.1	-36.3	8.3	-15.5	7.4	-16.9	1.2	50	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	27.8	17.5	26.0	16.8	24.3	16.1	19.0	25.1	18.0	23.7	17.1	22.5	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.3	21.0	15.8	11.6	20.1	14.9	10.9	19.4	54.9	25.4	51.9	23.7	49.1	22.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.5	7.4	6.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-42.5	31.0	2.4	1.7	-44.2	32.3	-45.6	33.2	-47.0	34.2	-48.7	35.4
			-42.2	20.6	2.2	1.2	-43.8	21.4	-45.1	22.1	-46.3	22.8	-47.9	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.0	-20.9	-18.3	-11.6	-0.8	7.9	14.6	17.4	15.1	9.3	1.3	-9.7	-17.7	(6)
(7)		DBStd	15.02	9.06	8.03	9.03	7.01	5.46	4.07	3.24	3.74	4.55	4.69	6.94	8.71	(7)
(8)		HDD10.0	4652	958	793	669	327	105	9	0	5	67	271	591	857	(8)
(9)		HDD18.3	7116	1216	1027	928	574	327	122	56	111	272	527	841	1116	(9)
(10)		CDD10.0	630	0	0	0	2	38	148	230	164	45	2	0	0	(10)
(11)		CDD18.3	52	0	0	0	0	2	11	28	11	1	0	0	0	(11)
(12)		CDH23.3	578	0	0	0	0	29	132	282	123	12	0	0	0	(12)
(13)		CDH26.7	102	0	0	0	0	4	22	53	21	2	0	0	0	(13)
(14)	Wind	WSAvg	3.0	2.6	2.7	3.0	3.5	3.6	3.3	3.0	2.9	2.9	2.8	2.7	2.6	(14)
(15)	Precipitation	PrecAvg	417	22	15	19	21	30	48	67	49	41	36	29	24	(15)
(16)		PrecMax	479	39	34	36	59	66	128	116	81	100	102	85	42	(16)
(17)		PrecMin	302	8	1	6	1	9	10	24	21	11	2	9	8	(17)
(18)		PrecStd	56	8	7	8	16	15	28	27	19	23	23	17	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.0	9.5	18.5	26.6	29.2	30.5	29.1	25.0	16.9	7.2	2.7	(19)
(20)			MCWB	-0.1	1.0	3.8	9.3	14.3	17.4	19.0	18.8	16.3	10.7	3.9	0.5	(20)
(21)		2%	DB	-1.8	-1.3	6.1	14.8	23.1	26.6	28.1	26.4	21.4	13.0	3.3	-0.8	(21)
(22)			MCWB	-2.9	-3.0	1.9	7.2	12.3	16.1	17.9	17.7	14.5	8.4	1.3	-2.0	(22)
(23)		5%	DB	-5.3	-4.2	3.6	11.9	20.2	24.7	26.5	24.4	18.9	10.3	1.2	-3.4	(23)
(24)			MCWB	-6.0	-5.4	0.6	5.6	11.1	15.4	17.4	16.7	13.2	6.6	0.0	-4.2	(24)
(25)		10%	DB	-8.3	-6.8	1.0	9.2	17.6	22.7	24.7	22.5	16.7	8.2	-0.3	-5.9	(25)
(26)			MCWB	-8.8	-7.7	-1.1	4.3	9.9	14.8	16.8	15.9	12.1	5.6	-1.2	-6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	1.0	4.3	9.7	15.4	19.0	20.5	20.5	17.0	11.6	4.0	0.7	(27)
(28)			MCDWB	1.3	3.1	9.1	17.3	24.3	26.1	27.5	26.6	23.8	15.7	6.7	2.5	(28)
(29)		2%	WB	-2.8	-2.8	2.2	7.5	13.2	17.7	19.2	18.7	15.2	8.6	1.6	-2.0	(29)
(30)			MCDWB	-1.8	-1.6	5.6	14.0	21.0	23.9	25.5	24.2	20.2	12.0	2.9	-1.0	(30)
(31)		5%	WB	-5.9	-5.3	0.7	5.9	11.9	16.5	18.4	17.6	13.8	7.0	0.2	-4.2	(31)
(32)			MCDWB	-5.2	-4.1	3.5	11.3	19.2	22.5	24.2	22.7	17.5	9.8	1.1	-3.5	(32)
(33)		10%	WB	-8.8	-7.7	-1.1	4.4	10.4	15.5	17.5	16.6	12.7	5.7	-1.2	-6.5	(33)
(34)			MCDWB	-8.2	-6.8	1.1	9.1	16.9	21.1	23.0	21.3	16.1	8.1	-0.3	-5.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	11.2	12.6	11.8	12.2	12.0	11.5	11.2	10.3	7.6	7.6	8.7	(35)
(36)			MCDBR	10.6	12.3	12.9	15.0	16.4	14.6	14.2	14.3	13.7	11.5	7.9	9.8	(36)
(37)			MCWBR	9.9	11.0	9.5	9.1	8.0	6.7	6.3	7.1	7.8	7.3	6.5	8.9	(37)
(38)		5% WB	MCDBR	10.6	12.3	12.7	14.3	14.8	12.5	12.3	12.7	12.0	10.7	7.5	9.3	(38)
(39)			MCWBR	9.9	11.0	9.5	8.9	7.6	6.0	6.7	7.1	7.0	6.3	8.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.212	0.238	0.275	0.322	0.343	0.369	0.389	0.353	0.319	0.282	0.233	0.190	(40)
(41)		taud		2.226	2.318	2.296	2.362	2.424	2.356	2.366	2.441	2.547	2.555	2.289	2.122	(41)
(42)		Ebn at Noon		664	825	878	888	889	866	837	847	830	769	622	554	(42)
(43)		Edh at Noon		42	64	90	102	103	112	109	94	72	53	39	31	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.48	1.35	2.98	4.76	5.79	6.03	5.55	4.45	2.88	1.37	0.51	0.29	(44)
(45)		RadStd		0.06	0.10	0.16	0.29	0.49	0.42	0.33	0.25	0.26	0.16	0.07	0.03	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) +1.92	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page